THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY
(Approved by AICTE, Govt. of Maharashtra & Affiliated to University of Mumbai)
A - Block, Thakur Educational Campus, Shyamnarayan Thakur Marg,
Thakur Village, Kandivali (East), Mumbai - 400 101.

TCET/EXAM/FRM/IP-03/06 REVISION: B

IN-SEMESTER EXAMINATION-II

F.E./F.T. (Semester-I)

SUBJECT – Mathematics-I

Branch: AIML, E&TC, E&CS, IT, MECH, MME **Date: 22/12/2025**
Div.: ALL **Timing: 4:00 PM to 05:00 PM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Express $\begin{bmatrix} 0 & -4 \\ 4 & 0 \end{bmatrix}$ as sum of Hermitian and Skew Hermitian matrices.	2	4	R
	b.	Check matrix $A = \begin{bmatrix} 5i & 7 \\ 7 & 2 \end{bmatrix}$ is Hermitian or not.	2	4	U
	c.	Find A^{-1} if $A = \begin{bmatrix} cosx & sinx \\ -sinx & cosx \end{bmatrix}$ is orthogonal matrix.	2	4	U
	d.	Evaluate $d^2\ (1/x)$ by taking '2' as the interval of differencing.	2	5	U
	e.	Show that $E^{-1} = 1 - \nabla$.	2	5	R
	f.	Find the value of $\left[\frac{1}{9} \quad \frac{1}{3}\right]$.	2	6	R
	g.	Evaluate $\int x^3(1-x)^4 dx$.	2	6	R
Q.2	a.	Investigate for what values of λ and μ the equations $2x + 3y + 5z = 9,\quad 7x + 3y - 2z = 8\quad 2x + 3y + \lambda z = \mu$ have (i) no solution (ii) unique solution (iii) many solutions.	5	4	A
		OR			
	b.	Evaluate the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ by reducing to normal form.	5	4	A

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<td>Q.3</td>
<td>a.</td>
<td>For a certain town the population was given below:
<table>
<tr><td>Year</td><td>1931</td><td>1941</td><td>1951</td><td>1961</td></tr>
<tr><td>Population in thousand</td><td>172</td><td>171.7</td><td>157.2</td><td>183.9</td></tr>
</table>
Find the population for 1946.</td>
<td>5</td>
<td>5</td>
<td>A</td>
</tr>
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<td></td><td></td><td>OR</td><td></td><td></td><td></td>
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<td></td>
<td>b.</td>
<td>Evaluate $\int_0^{\pi/6} \cos^6 3\theta \sin^2 6\theta \, d\theta$</td>
<td>5</td>
<td>6</td>
<td>A</td>
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